

CHAPTER 4

SYSTEM REQUIREMENTS SPECIFICATION

4.1 HARDWARE REQUIREMENTS

- System : Intel i5
- GPU : Minimum NVIDIA 4 GB VRAM
- Server : 2-core CPU, 2-4 GB RAM
- Input Devices : Any modern computer with a web browser
- Ram : 8 GB

4.1.1 CPU -INTEL CORE i5

Intel Core is a brand name that Intel uses for various mid-range to high-end consumer and business microprocessors. As of 2015 the current line-up of Core processors included the Intel Core i7, Intel Core i5, and Intel Core i3. The CPU is the main processing unit of the system that executes all the computational and logical operations required for the project. In this EEG-based Depression Screening system, the CPU is responsible for performing data preprocessing tasks such as filtering EEG signals, feature extraction, and managing the workflow of the hybrid neural network model during training and testing. It also handles backend operations of the web application, ensuring smooth communication between the user interface and the machine learning model.

4.1.2 RAM

Ram acts as a temporary storage area that provides fast access to data and instructions while the program is running. RAM is used to store EEG datasets, intermediate results, and model parameters during model training and testing. Sufficient RAM ensures that the model can process large amounts of EEG data efficiently without performance issues or lag, thereby improving overall system speed and stability. RAM also supports parallel data handling, which allows multiple operations such as data preprocessing, feature extraction, and model computation to run simultaneously. Additionally, having higher RAM capacity minimizes the

need for frequent disk access, reducing latency and enhancing the real-time performance of EEG-based depression detection systems.

4.1.3 SERVER

The server is used for deploying the web application so that it can be accessed remotely by end users through a web browser. It hosts the trained machine learning model and handles user requests such as uploading EEG data and generating depression screening results. The server ensures real-time communication between the user interface and the backend model, providing a reliable and accessible platform for EEG-based depression prediction.

4.2 SOFTWARE REQUIREMENTS

➤ Operating System	:	Windows11
➤ Coding Language	:	Python 3.10
➤ Frontend Language	:	HTML5, CSS3, JavaScript (ES6+)
➤ Backend Language	:	Django, SQLite
➤ Tools	:	Keras, TensorFlow ,SciPy, MNE-Python
➤ Visualization	:	Chart.js, Matplotlib , Seaborn

4.2.1 PYTHON

Python 3.10 serves as the core programming language for the EEG-based Depression Screening project. It provides a flexible and efficient environment for implementing every stage of the system — from EEG data preprocessing to deep learning model training and web deployment. Python’s vast ecosystem of scientific and AI libraries makes it ideal for handling biomedical signal processing and machine learning tasks.

In the DepHNN Analyzer, Python is used to build and train the hybrid CNN-LSTM model, process EEG signals using libraries like MNE, NumPy, and SciPy, and perform machine learning operations such as data splitting and evaluation with Scikit-learn. It also powers the Django web framework, enabling user authentication, model integration, and server-side logic. Additionally, visualization libraries like Matplotlib and Seaborn are used within Python to generate plots for model performance analysis.

4.2.2 TOOLS

➤ **TensorFlow & Keras:**

- The core deep learning framework used to build, train, and save our CNN-LSTM model.

➤ **MNE-Python:**

- The primary library for biomedical signal processing. We used it to load .edf files, apply filters, run Independent Components Analysis (ICA), and segment data.

➤ **Scikit-learn:**

- Used for data splitting (train/test split) and for calculating evaluation metrics (classification report, confusion matrix).

➤ **NumPy:**

- The fundamental library for all numerical operations and handling our data arrays.

➤ **Pandas:**

- Used for data manipulation (though our final pipeline relied mostly on NumPy).

➤ **SciPy:**

- A core library required by MNE for its scientific computing functions.

4.2.3 VISUALIZATION

➤ **JavaScript:**

- The JavaScript library used to render the final prediction (healthy vs. depressed) as an interactive donut chart.

➤ **Matplotlib & Seaborn:**

- Python libraries used in our evaluate_model.py script to generate the static confusion matrix and training history plots for our report.

CHAPTER 5

SYSTEM DESIGN